PSX FIX DROP RASH Format

Version 1.00

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1. Overview

NASDAQ OMX PSX accepts limit orders from subscribers and executes matching orders when possible. Non- matching orders may be added to the NASDAQ OMX PSX Limit Order Book, a database of available limit orders, where they wait to be matched in price-time priority or routed out to an external destination for execution.

FIX DROP is an application that delivers real-time information about activity that takes place on NASDAQ OMX PSX in FIX format. Each FIX DROP account is configured to transmit information concerning orders entered by one or more NASDAQ OMX PSX subscriber firms.

Each FIX DROP host can be configured to send a message anytime an order is entered, canceled, executed or broken – or any combination of these events.

FIX DROP does not provide the ability to enter orders into NASDAQ OMX PSX. To enter orders into NASDAQ OMX PSX, subscribers should use one of our connectivity protocols. Available separately.

1.1. Architecture

NASDAQ OMX PSX's RASH FIX DROP format is transmitted using the Financial Information Exchange (FIX) protocol.

Subscribers are required to log in to the FIX DROP application using a standard FIX logon message. Once logged in, the subscriber will need to send only a heartbeat message to keep the session active.

In the case where a subscriber loses a connection to the FIX DROP host, they can reconnect in either NASDAQ OMX PSX's primary or backup data center. When reconnecting, the subscriber should re-login to the session using the next transmitted sequence numbers. In the event that the sequence numbers are out of synch, the subscriber should send a resend request to retransmitted any potentially lost data.

1.2. Service Bureau Configuration

A single FIX DROP host can deliver information for one or more subscribers, allowing a service bureau configuration. In this case, each subscriber listed on the FIX DROP account must sign a Service Bureau Agreement providing authorization for the Service Bureau to receive the drop copies.

2. FIX DROP Configuration

Each FIX DROP port can be configured to include or exclude trades based on the subscriber's MMID, INET port and SenderSubID. Subscribers may receive drop copies for all their execution or limit it to specific ports. Configuration requests for existing ports should be directed to NASDAQ OMX PSX Subscriber Services at subscriber@nasdaq.com.

3. FIX Protocol Messages

3.1. Message Header

Message Header

Tag	Field Name	Req'd	Comments
8	BeginString	Y	FIX 4.0, 4.1 or 4.2
9	BodyLength	Y	Must be second field in message
35	MessageType	Y	
128	DeliverToSubID	Y	NASDAQ OMX PSX Account
49	SenderCompID	Y	NASDAQ OMX PSX
56	TargetCompID	Y	As assigned by NASDAQ OMX PSX
43	PossDupFlag	N	Always required for retransmitted messages.
34	MsgSeqNum	Y	
52	SendingTime	Y	Time of message transmission
122	OrigSendingTime	Y	Original time of message transmission
50	SenderSubID	N	NASDAQ OMX PSX
57	TargetSubID	N	This value will contain the SenderSubID received in order entry message. For trades entered via the Inet Terminal, this field will contain the Inet Terminal ID. If no SenderSubID is sent in the order entry message, it will default to the first four character of the NASDAQ OMX PSX account name.

3.2. Logon Message

Logon Message

Tag	Field Name	Req'd	Comments
	Standard Header	Y	MsgType = A
98	EncryptMethod	Y	Not Supported
108	HeartBtInt	Y	
141	ResetSeqNumFlag	Y	Indicates both sides of a FIX session should reset sequence numbers
	Standard Trailer	Y	

3.3. Execution Report

Execution Report

Tag	Field Name	Req'd	Comments
	Standard Header	Y	MsgType = 8
11	ClOrdID	Y	Client order ID as assigned in new order message
37	OrderID	Y	OrderID, as assigned by INET, is required to be unique for each chain of orders.
17	ExecID	Y	Identifier of execution message as assigned by broker (will be 0 (zero) for ExecTransType=3 (Status)). INET ATS assigns the ExecID based on the OUCH match number of the transaction. If you cross yourself on a trade the buy and sell orders will have the same ExecID.

Execution Report

Tag	Field Name	Req'd	Comments
20	ExecTransType	Y	Identifies transaction type Valid values: 0 = New 1 = Cancel Trade breaks will be sent with an ExecTransType of '1'. NASDAQ OMX PSX does not support trade corrections.
150	ExecType	Y	Describes the Specific Execution Report Valid Values: 0 = New 1 = Partial Fill 2= Filled 4 = Canceled 5 = Replaced 6 = Pending Cancel 8 = Rejected
39	OrdStatus	Y	Describes the current state of a CHAIN of orders, same scope as OrderQty, CumQty, LeavesQty, and AvgPx Valid Values: 0 = New 1 = Partial Fill 2= Filled 4 = Canceled 5 = Replaced 6 = Pending Cancel 8 = Rejected
109	ClientID	N	Executing Firm's MMID
55	Symbol	Y	Ticker Symbol
54	Side	Y	Side of order Valid values: 1 = Buy 2 = Sell 5 = Short Sell 6 = Short Sell Exempt
20107	GroupId	N	Numeric XXXXX, max 32767
38	OrderQty	Y	
32	LastShares	Y	Quantity of shares bought/sold on this (last) fill.
31	LastPx	Y	Price of this (last) fill.
151	LeavesQty	N	Amount of shares open for further execution.
14	CumQty	Y	Currently executed shares for chain of orders.
6	AvgPx	Y	Calculated average price of all fills on this order.
18	ExecInst	Y	Order Handling Instructions
44	Price	N	As specified in the NewOrder
40	Order Type	N	Valid Values: 1 = Market 2 = Limit 3 = Cross

Execution Report

Tag	Field Name	Req'd	Comments
47	Rule80A	N	Order Capacity Valid Values: A = Agent E= Registered Market Maker Trade P = Principal
29	LastCapacity	N	Broker Capacity In Order Execution Valid Values: 1 = Agent 2 = Cross as agent 3 = Cross as principal 4 = Principal
76	Exec Broker	N	INET
59	TimeInForce	N	Valid Values: 0 = Day 2 = At the Opening 3 = Immediate or Cancel 5 = Good Till Crossing (GTX) 7 = At the Close R = Re-routed every 4 minutes (applies to certain routing strategies) C = Market Hours Good-Til-Cancel (GTC) M = System Hours Good-Til-Cancel (GTC)
58	Text	N	Text Message
75	TradeDate	N	Date of Execution
60	TransactTime	N	Time of execution/order creation
9882	LiquidityFlag	N	Will be returned within all execution reports that contain a partial or full fill. A = Added Liquidity R = Reduced Liquidity V = Displayed added liquidity with original order size of greater than or equal to 2,000 shares. Only applicable to symbols that are set to the pro rata execution algorithm. X = Routed D = DOT Routed F = Opening Trade (on NYSE) G = On-Close order (on NYSE) O = Opening Cross M = Opening Cross (imbalance-only) C = Closing Cross L = Closing Cross (imbalance-only) H = Halt/IPO Cross K = Halt Cross J = Non-displayed adding liquidity S = Odd Lot Executions (on NYSE) U = Added Liquidity (on NYSE) B = Routed to OMX PSX E = NYSE Other P = Routed to PSX T = Opening Trade (on ARCA) Z = On-Close order (on ARCA) m = Removed liquidity at a midpoint k = Added liquidity via a midpoint order
1	Account		May be user field, otherwise not present
21002	Locate Broker	N	Will only be populated if a single broker is designated per session in the EQRC Shares Located rule.
	Standard Trailer	Y	

3.4. Restatement Message

This message will tell the user the amount their orders were reduced down to as a result of a partially cancelled quantity.

Restatement

Tag	Field Name	Req'd	Comments
	Standard Header	Y	MsgType = 8
11	ClOrdID	Y	Client order ID as assigned in new order message
37	OrderID	Y	OrderID, as assigned by INET, is required to be unique for each chain of orders.
17	ExecID	Y	Identifier of execution message as assigned by broker (will be 0 (zero) for ExecTransType=3 (Status)). INET ATS assigns the ExecID based on the OUCH match number of the transaction. If you cross yourself on a trade the buy and sell orders will have the same ExecID.
20	ExecTransType	Y	Identifies transaction type Valid values: 0 = New 1 = Cancel Trade breaks will be sent with an ExecTransType of '1'. NASDAQ OMX PSX does not support trade corrections.
150	ExecType	Y	Describes the type of execution report. For restatement, this value will always be: D = Restated
39	OrdStatus	Y	Describes the current state of a CHAIN of orders, same scope as OrderQty, CumQty, LeavesQty, and AvgPx Valid Values: 0 = New 1 = Partial Fill 2= Filled 4 = Canceled 5 = Replaced 6 = Pending Cancel 8 = Rejected
378	ExecRestatementReason	Y	For restatement, this value will always be: 5 = Partial decline of OrderQty
109	ClientID	N	Executing Firm's MMID
55	Symbol	Y	Ticker Symbol
54	Side	Y	Side of order Valid values: 1 = Buy 2 = Sell 5 = Short Sell 6 = Short Sell Exempt
20107	GroupId	N	Numeric XXXXX, max 32767
38	OrderQty	Y	
31	LastPx	Y	Price of this (last) fill.
14	CumQty	Y	Currently executed shares for chain of orders.
111	MaxFloow	N	As specified in the NewOrder
44	Price	N	As specified in the NewOrder
40	Order Type	N	Valid Values: 1 = Market 2 = Limit

Restatement

Tag	Field Name	Req'd	Comments
47	Rule80A	N	Order Capacity Valid Values: A = Agent E= Registered Market Maker Trade P = Principal
29	LastCapacity	N	Broker Capacity In Order Execution Valid Values: 1 = Agent 2 = Cross as agent 3 = Cross as principal 4 = Principal
76	Exec Broker	N	INET
59	TimeInForce	N	Valid Values: 0 = Day 2 = At the Opening 3 = Immediate or Cancel 5 = Good Till Crossing (GTX) 7 = At the Close
	Standard Trailer	Y	

4. Revision History

Revision #	Date	Notes
1.0	06/12/2014	Updated formatting of the document
1.0	04/02/2015	Greyed out liquidity flag values for crosses Removed liquidity flag values “I”, ”9”
1.0	06/04/2015	Changed wording of liquidity flag values “O”, “M”, “C”, “L”, “H”, and “K” to make consistent with all specifications
1.0	06/08/2015	Changed wording of liquidity flag “V”

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